

- A. A.A We can see that α is smooth because each of its component functions are polynomials, which are smooth. We show that α is regular by showing that $\dot{\alpha}(u) \neq 0$ for all u . We see that $\dot{\alpha} = (1, 2u)$, which can never be the zero vector. We can then use $\dot{\alpha}$ to compute the unit tangent vector field $\mathbf{t}$ (now known as $\mathbf{T}$). We compute

$$\mathbf{t}(u) = \frac{\dot{\alpha}}{|\dot{\alpha}|} = \frac{\begin{bmatrix} 1 \\ 2u^2 \end{bmatrix}}{\sqrt{1 + 4u^2}}.$$

- A.B We can see that α is smooth because each of its component functions are polynomials, which are smooth. We show that α is not regular by showing that $\dot{\alpha} = 0$ for some u . We see that $\dot{\alpha}(u) = (3u^2, 6u^5)$, which is the zero vector when $u = 0$. Therefore, because α is not regular, it is not possible to compute the unit tangent vector field $\mathbf{t}$ because it is not defined when $\dot{\alpha} = \mathbf{0}$.

- B. B.A The speed of α is given by $\sqrt{a^2 \cos^2(u) + b^2 \sin^2(u)}$. To find the minima and maxima of the speed, we can instead optimize the square of the speed. We see that $|\dot{\alpha}|^2 = a^2 \cos^2(u) + b^2 \sin^2(u)$. We differentiate to find the minima and maxima. We compute that

$$\frac{d}{du} |\dot{\alpha}|^2 = 2a^2 \sin(u) \cos(u) - 2b^2 \sin(u) \cos(u) = 2 \sin(u) \cos(u) (a^2 - b^2).$$

By observation we see that this equals 0 whenever $a = b$ (which never happens because $a > b > 0$), and when $\sin(u) = 0$ or $\cos(u) = 0$. This occurs $\frac{\pi}{2}k$ for $k \in \mathbb{Z}$, making $\sin$ and $\cos$ equal to 0 at alternating k . At πk , the speed is

$$|\dot{\alpha}| = \sqrt{a^2 \cos^2(\pi k)} = a.$$

Similarly, at $\pi k + \frac{\pi}{2}$, the speed is

$$|\dot{\alpha}| = \sqrt{b^2 \sin^2(\pi k + \frac{\pi}{2})} = b.$$

By assumption, $a > b$, and therefore the speed is maximized at all points πk and minimized at all points $\pi k + \frac{\pi}{2}$ for $k \in \mathbb{Z}$.

- B.B No. The parameterization u does not need to be related to t , and therefore B.A only tells us the speed relative the u , not the actual time t .

- C. C.A We compute the angle θ between α and $\dot{\alpha}$ by computing $\arccos(\frac{\alpha \cdot \dot{\alpha}}{|\alpha| |\dot{\alpha}|})$. Let $a = ce^{\lambda u} \cos(u)$ and $b = ce^{\lambda u} \sin(u)$, noting that $a^2 + b^2 = c^2 e^{2\lambda u}$. Then $\alpha = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\dot{\alpha} = \begin{bmatrix} \lambda a - b \\ \lambda b + a \end{bmatrix}$. We compute that $|\alpha| = \sqrt{a^2 + b^2} = ce^{\lambda u}$. We compute that $|\dot{\alpha}| = \sqrt{(\lambda a - b)^2 + (\lambda b + a)^2}$. We see that the radicand simplifies to $\lambda^2 a^2 + \lambda^2 b^2 + a^2 + b^2$, which is equivalent to $(a^2 + b^2)(\lambda^2 + 1)$. Plugging back in, we get that

$$|\dot{\alpha}| = \sqrt{c^2 e^{2\lambda u} (\lambda^2 + 1)} = ce^{\lambda u} \sqrt{\lambda^2 + 1}.$$

We also compute $\boldsymbol{\alpha} \cdot \dot{\boldsymbol{\alpha}}$ elementwise. We see that

$$\boldsymbol{\alpha} \cdot \dot{\boldsymbol{\alpha}} = \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} \lambda a - b \\ \lambda b + a \end{bmatrix} = a(\lambda a - b) + b(\lambda b + a) = \lambda c^2 e^{2\lambda u}.$$

Putting it all together, we see that

$$\arccos\left(\frac{\boldsymbol{\alpha} \cdot \dot{\boldsymbol{\alpha}}}{\|\boldsymbol{\alpha}\|\|\dot{\boldsymbol{\alpha}}\|}\right) = \arccos\left(\frac{\lambda c^2 e^{2\lambda u}}{(c e^{\lambda u})(c e^{\lambda u} \sqrt{\lambda^2 + 1})}\right) = \arccos\left(\frac{\lambda}{\sqrt{\lambda^2 + 1}}\right),$$

which is θ and constant.

C.B Recall that the arc length of a parameterized curve is given as a function of u for a fixed u_0 by $s(u) = \int_{u_0}^u \|\dot{\boldsymbol{\alpha}}(v)\| dv$. Fix $u_0 = 0$. For this particular $\boldsymbol{\alpha}$, this amounts to

$$s(u) = c\sqrt{\lambda^2 + 1} \int_{u_0}^u e^{\lambda v} dv = \frac{c\sqrt{\lambda^2 + 1}}{\lambda} (e^{\lambda u} - 1).$$

We can solve for u to find the inverse $u(s)$. We compute that

$$u(s) = \frac{1}{\lambda} \ln\left(\frac{\lambda s}{c\sqrt{\lambda^2 + 1}} + 1\right).$$

Using $u(s)$ to parameterize $\boldsymbol{\alpha}(\mathbf{u})$, we see that

$$\boldsymbol{\alpha}(\mathbf{u}(s)) = c\left(\frac{\lambda s}{c\sqrt{\lambda^2 + 1}} + 1\right) \begin{bmatrix} \cos\left(\frac{1}{\lambda} \ln\left(\frac{\lambda s}{c\sqrt{\lambda^2 + 1}} + 1\right)\right) \\ \sin\left(\frac{1}{\lambda} \ln\left(\frac{\lambda s}{c\sqrt{\lambda^2 + 1}} + 1\right)\right) \end{bmatrix}.$$